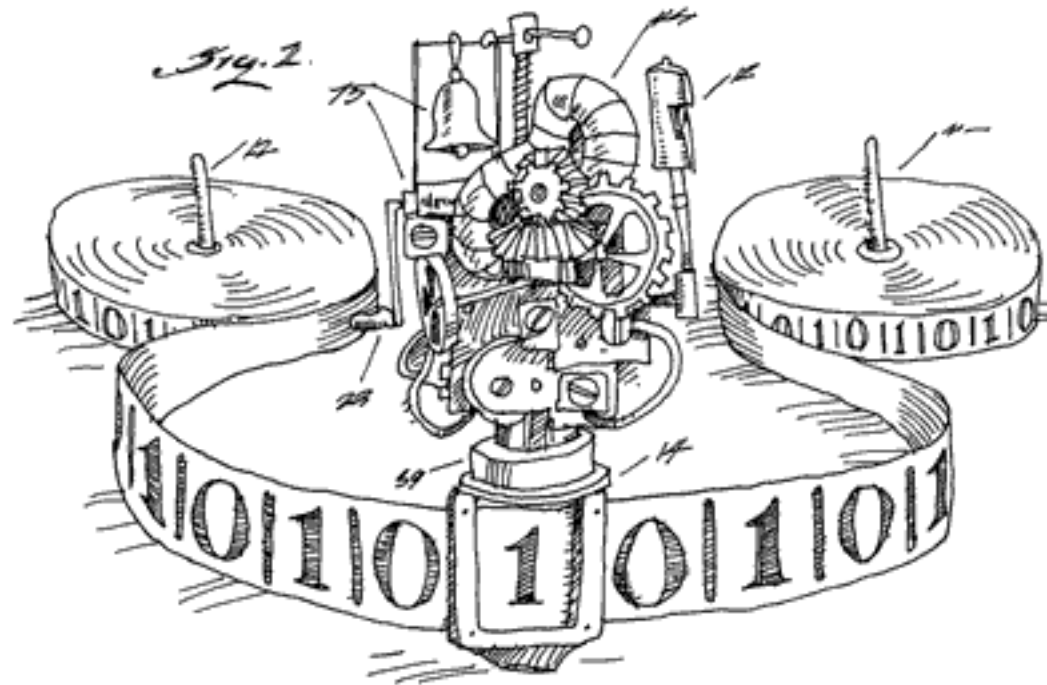


CSC4010: Introduction to Theoretical Computer Science



Tutorial 1

Recap: Cardinality of Infinite Sets

Definition: Two (infinite/finite) sets A, B have the same cardinality $|A| = |B|$ if and only if there is a bijection from A to B .

Claim: For $\mathbb{Z}^+ = \{1, 2, \dots\}$ and $\mathbb{N} = \{0, 1, 2, \dots\}$, $|\mathbb{Z}^+| = |\mathbb{N}|$.

Equivalent Definition: Two (infinite/finite) sets A, B have the same cardinality $|A| = |B|$ if and only if

- a) There is a one-one function $f : A \rightarrow B$, and
- b) there is a one-one function from $g : B \rightarrow A$.

Example 2: $|\mathbb{N}| = |\mathbb{Z}|$

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- a) There is a one-one function $f : A \rightarrow B$, and
- b) there is a one-one function from $g : B \rightarrow A$.

- **Recall:** $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ and $\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}$.
- **(Easy Case):** $f : \mathbb{N} \rightarrow \mathbb{Z}$ that is one-one.

Define $f(x) = x$

- **(Hard Case):** $g : \mathbb{Z} \rightarrow \mathbb{N}$ that is one-one.

$$\begin{array}{ccccccc} \mathbb{Z} : & 0, & 1, & -1, & 2, & -2, & 3, & -3, & \dots \\ & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow & \downarrow & \\ \mathbb{N} : & 0, & 1, & & 2, & 3, & 4, & 5, & 6, & \dots \end{array}$$

$$g(x) = \begin{cases} 2x - 1 & \text{if } x \geq 0, \\ -2x & \text{if } x < 0. \end{cases}$$

Example 2: $|\mathbb{N}| = |\mathbb{Z}|$

$\mathbb{Z} : 0, 1, -1, 2, -2, 3, -3, \dots$

$\downarrow \downarrow \quad \downarrow \downarrow \quad \downarrow \downarrow \quad \downarrow$
 $\mathbb{N} : 0, 1, \quad 2, 3, \quad 4, 5 \quad 6$

- **(Hard Case):** $g : \mathbb{Z} \rightarrow \mathbb{N}$ that is one-one. This is equivalent to listing out all the infinitely many element of $\mathbb{Z}$ such that each $x \in \mathbb{Z}$ has a finite position on this list.
- **Another way to list:** List $\mathbb{Z}^+ \cup \{0\}$ first, and then $\mathbb{Z}^-$. But $\mathbb{Z}^+ \cup \{0\}$ is infinite! So, we do not have a finite index for -1 .

Definitions

- **Definition:** A set is **countable** if $|A| \leq |\mathbb{N}|$. This is the same as saying there is a one-one function from A to $\mathbb{N}$.
- **Definition:** A set is **countably infinite** if $|A| = |\mathbb{N}|$.

Example 3: $\mathbb{N}^2$ is countably infinite

Equivalent Definition: Two (infinite/finite) sets A, B have the same cardinality $|A| = |B|$ if and only if

- a) There is a one-one function $f : A \rightarrow B$, and
- b) there is a one-one function from $g : B \rightarrow A$.

- **Recall:** $\mathbb{N}^2 = \{(x, y) \mid x, y \in \mathbb{N}\}$.
- **(Easy Case):** $f : \mathbb{N} \rightarrow \mathbb{N}^2$ that is one-one.
Define $f(x) = (x, 0)$.
- **(Hard Case):** $g : \mathbb{N}^2 \rightarrow \mathbb{N}$ that is one-one.
Define $g((x, y)) = 2^x 3^y$

Example 4: $\mathbb{N}^3$ is countably infinite

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- a) There is a one-one function $f : A \rightarrow B$, and
- b) there is a one-one function from $g : B \rightarrow A$.

- **Recall:** $\mathbb{N}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{N}\}$.
- **(Easy Case):** $f : \mathbb{N} \rightarrow \mathbb{N}^3$ that is one-one.
Define $f(x) = (x, 0, 0)$.
- **(Hard Case):** $g : \mathbb{N}^3 \rightarrow \mathbb{N}$ that is one-one.
Define $g((x, y, z)) = 2^x 3^y 5^z$

Example 5: $\mathbb{N}^k$ is countably infinite

Claim: For any $k \in \mathbb{Z}^+$, $\mathbb{N}^k$ is countably infinite.

- **(Hard Case):** $g : \mathbb{N}^k \rightarrow \mathbb{N}$ that is one-one.

$$\text{Define } g((x_1, x_2, \dots, x_k)) = p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}$$

Where p_i is the i th prime number.

Example 6: The set of all finite binary strings

- Let $\{0,1\}^k$ be the collection of all strings of length k .
- For example: $\{0,1\}^2 = \{00,01,10,11\}$.
- $\{0,1\}^0$ is the empty string, denoted ε .
- $\{0,1\}^* = \cup_{k \geq 0} \{0,1\}^k$ be the collection of all finite strings.

Claim: $\{0,1\}^*$ is countably infinite.

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Claim: $\{0,1\}^*$ is countably infinite.

$f: \{0,1\}^* \rightarrow \mathbb{N}$ that is one-one.

- First order every string by length.
- Within each length, order by dictionary order.

Example 7: $\mathbb{Q}$ is countably infinite

Claim: $|\mathbb{Q}| = |\mathbb{N}|$ is countably infinite.

- **Recall:** $\mathbb{Q} = \left\{ \frac{x}{y} \mid x, y \in \mathbb{N}^+ \right\}$.
- Instead of finding a one-one function, we will list out all the infinitely many element of $\mathbb{Q}$ such that each $\frac{x}{y} \in \mathbb{Q}$ has a finite position on this list.

1/1	1/2	1/3	1/4	...
2/1	2/2	2/3	2/4	...
3/1	3/2	3/3	3/4	...
4/1	4/2	4/3	4/4	...
5/1	5/2	5/3	5/4	...
⋮	⋮	⋮	⋮	

Convert this matrix into a list!

1) Read off row by row.

2) Read off column by column.

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2/1	2/2	2/3	2/4	...
3/1	3/2	3/3	3/4	...
4/1	4/2	4/3	4/4	...
5/1	5/2	5/3	5/4	...
⋮	⋮	⋮	⋮	

Convert this matrix into a list!

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2) Read off column by column.



Next Class: Is Everything Countable?